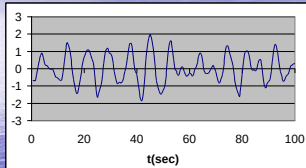


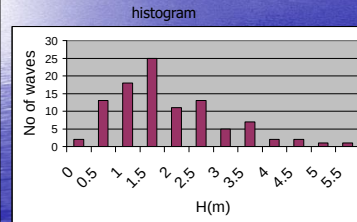
Wave Climate Statistics



Short term analysis
One storm data
H_i, T_i
H_s, T_s
Characteristic wave parameters
Wave height distribution (period)
Spectrum analysis

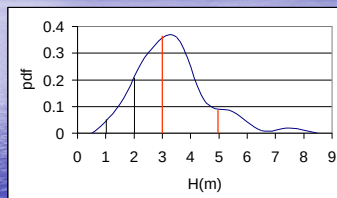
Many storms belong to several years: Long term wave data.
Each wave in the data series summarizes a short term wave condition: H_s, T_s

Wave height distribution



	H(m)	no of waves
0-0.5	0.25	2
0.5-1	0.75	13
1-1.5	1.25	18
1.5-2	1.75	25
2-2.5	2.25	11
2.5-3	2.75	13
3-3.5	3.25	5
3.5-4	3.75	7
4-4.5	4.25	2
4.5-5	4.75	2
5-5.5	5.25	1
5.5-6	5.75	1
		100

Wave height distribution



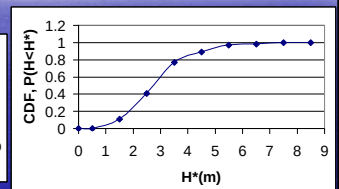
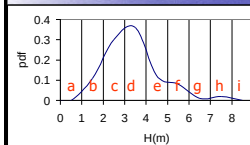
$P(H < 3m) = ?$
 $P(H > 3m) = ?$
 $P(3m < H < 5m) = ?$
 $P(H < 9m) = ?$

Area under pdf should be 1

Probability density function = pdf = $p(x) = \text{Probability } P(x)/dx$

$$p(x) = p(H) = \frac{n}{N} \frac{1}{\Delta H}$$

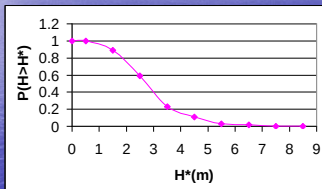
Wave height distribution



CDF: Cumulative Distribution Function

CDF is obtained by integrating pdf to yield the probability that the individual wave Height H, is not higher than a specified wave height

Wave height distribution



The probability of exceedance, $Q(x)=1-P(x)$
the probability that the individual wave height H , is higher than a specified wave height

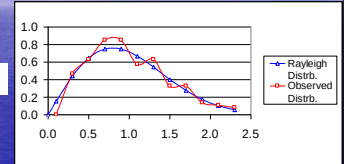
Wave height distribution

Rayleigh distribution
CDF:

$$P(H < H') = 1 - \exp(-\alpha^2 x^2)$$

$$x = H'/H^*$$

$$\alpha = \begin{cases} \sqrt{\pi}/2 & \text{if } H^* = H_{\text{mean}} \\ \sqrt{2} & \text{if } H^* = H_{1/3} \end{cases}$$



$$p(x) = \frac{dP}{dx} = 2\alpha^2 x \exp(-\alpha^2 x^2)$$

Wave height distribution

Rayleigh distribution, the prob. of exc.

$$Q(H \geq H') = 1 - P(H \leq H') = \exp(-\alpha^2 x^2)$$

$$x = H'/H^*$$

$$\alpha = \begin{cases} \sqrt{\pi}/2 & \text{if } H^* = H_{\text{mean}} \\ \sqrt{2} & \text{if } H^* = H_{1/3} \end{cases}$$

Given: $H_s = 4\text{m}$, $N = 200$ waves
question: prob of observing waves higher than 5m?, $H_{\text{max}} = ?$

$$Q(H > 5\text{m}) = e^{-2(5/4)} = 0.044$$

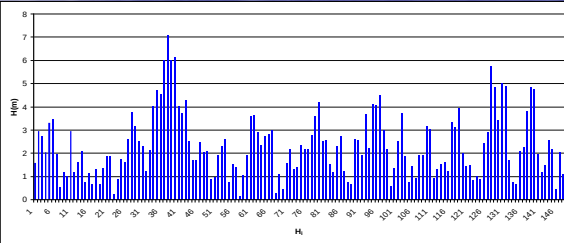
$$Q(H > H_{\text{max}}) = 1/N = e^{-2(H_{\text{max}}/H_s)}$$

$$H_{\text{max}} = H_s \sqrt{\ln N / 2} = 6.51\text{m}$$

Relation betw characteristic wave heights

- Using Rayleigh distrb
- $H_{1/10} = 1.27 H_s$
- $H_s = 1.6 H_{\text{mean}}$
- $H_{\text{max}} = (1.6-2) H_s$

Wave Climate Statistics



Wave climate analysis is carried out on long term wave height data; a series of observed or hindcast wave heights spanning years.

Long term wave analysis

We need:

- To organise wave height data
- To extrapolate the data set to extreme values of wave heights occurring at low probabilities of exceedence

Long term wave height analysis provides a distribution of prob of occurrence of wave height over several years

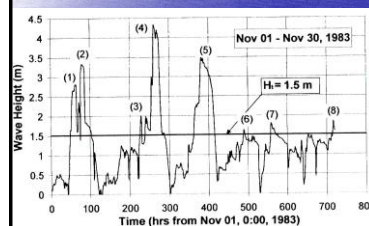
Long term wave analysis

wave height data?

- All data (total sample method)
- Grouped data obtained from a complete long-term data set (peaks over threshold method)
- Ordered data derived using a limited number of extreme values (annual maxima method)

Long term wave analysis

- Grouped data (storms $H > H_{\text{threshold}}$)
- Statistically independent



$H_{\text{threshold}} = ?$
Too high... number decreases
Too low... number decreases
Moderate level?

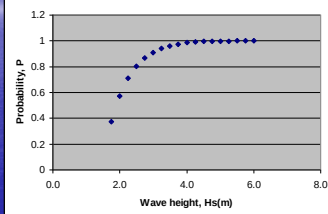
Minimum time interval
between successive peaks

2-4 days

Maximize the number of
Storm events

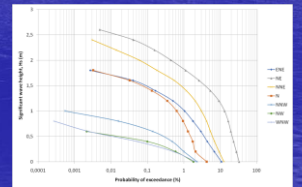
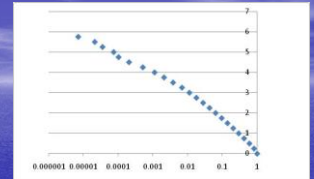
Long term wave analysis

Hs	N	ΣN	P
1.750	1019	1019	0.372169
2.000	549	1568	0.572681
2.250	382	1950	0.712199
2.500	254	2204	0.804967
2.750	174	2378	0.868517
3.000	113	2491	0.909788
3.250	81	2572	0.939372
3.500	60	2632	0.961286
3.750	40	2672	0.975895
4.000	27	2699	0.985756
4.250	19	2718	0.992696
4.500	10	2728	0.996348
4.750	4	2732	0.997809
5.000	2	2734	0.998339
5.250	1	2735	0.998904
5.500	2	2737	0.999335
5.750	0	2737	0.999635
6.000	1	2738	1
	2738		



By CDF it is possible to organise data but it is difficult to interpolate and extrapolate. CDF needs to be transformed into a straight line. We can fit the data to a theoretical probability Distribution or make the axis logarithmic

Hs-range	N	ΣN	P	Q
0.000	0.000	0.000	0.186246	0.813754
0.251	0.250	62679	0.119751	0.400459
0.501	0.750	60472	0.134801	0.429757
0.751	1.000	39781	0.195042	0.705766
1.001	1.250	25367	0.224958	0.795622
1.251	1.500	18153	0.247728	0.850086
1.501	1.750	12678	0.254271	0.904823
1.751	2.000	8894	0.264201	0.936372
2.001	2.250	5961	0.270262	0.957443
2.251	2.500	4143	0.274435	0.972119
2.501	2.750	2828	0.277281	0.981787
2.751	3.000	1895	0.279311	0.987125
3.001	3.250	1238	0.280368	0.993135
3.251	3.500	865	0.281231	0.996109
3.501	3.750	483	0.281716	0.997971
3.751	4.000	274	0.281960	0.998881
4.001	4.250	171	0.282181	0.999488
4.251	4.500	87	0.282348	0.999795
4.501	4.750	29	0.282477	0.999897
4.751	5.000	3	0.282585	0.999926
5.001	5.250	11	0.282696	0.999965
5.251	5.500	4	0.282800	0.999979
5.501	5.750	4	0.282844	0.999993
5.751	6.000	2	0.282908	1
		282908		



Extreme wave statistics

- Purpose: estimate an expected value of extreme event which would occur once in a long period of time
- Return period: average duration of time during which extreme events exceeding a certain threshold value would occur once.
- Return value: threshold value which defines a given return period.
- A 100 year wave height is a return value which would be exceeded once in every 100 years on the average.

Extreme wave statistics

$$Q = \frac{1}{\lambda T_r}$$

$$P = 1 - Q$$

P = Prob. of non-exceedance
Q = Prob. of exceedance

T_r = Return period in years
 λ = no. of events per year
mean rate = N_T/K

$$\lambda = 2738/39.4 \text{ years} = 78.45$$

Extreme value analysis from ordered data

- Developed specifically for the analysis of extreme values
- Limited no of max annual data or POT

year	Hs
1984	1.45
1985	1.71
1986	0.91
1987	2.99
1988	2.44
1989	4.9
1990	4.96
1991	2.33
1992	1.41
1993	1.6

Extreme value analysis from ordered data

- Sort in descending order

m	Hs
1	4.96
2	4.4
3	2.99
4	2.44
5	2.33
6	1.71
7	1.6
8	1.45
9	1.41
10	0.91

$$P = 1 - \frac{m - c_1}{N + c_2} \quad m = 1, 2, \dots, N$$

Commonly used values of c_1 and c_2 is 0 and 1 respectively. But it gives biased results! Values of c_1 and c_2 change according to used distribution

Ordered (extreme) statistics Gumbel distribution

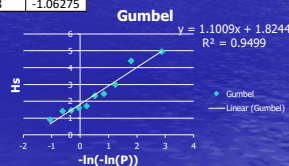
year	Hs(m)	ordered Hs	m	$P=1-(m-0.44/N+0.12)$	$-\ln(-\ln(P))$
1984	1.45	4.96	1	0.944664032	2.866004
1985	1.71	4.9	2	0.845849802	1.787289
1986	0.91	2.99	3	0.747035573	1.232227
1987	2.99	2.44	4	0.648221344	0.83581
1988	2.44	2.33	5	0.549407115	0.512635
1989	4.9	1.71	6	0.450592885	0.226661
1990	4.96	1.6	7	0.351778656	-0.04378
1991	2.33	1.45	8	0.252964427	-0.31809
1992	1.41	1.41	9	0.154150198	-0.62585
1993	1.6	0.91	10	0.055335968	-1.06275

$C1=0.44$
 $C2=0.12$ for Gumbel

$$P = \exp(-\exp(-\frac{H-\beta}{\alpha}))$$

$$-\ln(-\ln P) = \frac{H-\beta}{\alpha}$$

$$H = \alpha(-\ln(-\ln P)) + \beta$$



Risk Analysis

- The probabilities of occurrence and non-occurrence add up to unity for any year.

$$p + q = 1.0$$

- If a period of 2 years is considered:

$$(p + q)^2 = p^2 + 2pq + q^2 = 1.0$$

- For 3 years:

$$(p + q)^3 = p^3 + 3p^2q + 3pq^2 + q^3 = 1.0$$

- For n years:

$$(p + q)^n = \sum_{k=0}^n \binom{n}{k} p^{n-k} q^k = \sum_{k=0}^n \frac{n!}{k!(n-k)!} p^{n-k} q^k = 1.0$$

$$(p + q)^n = p^n + np^{n-1}q + \frac{n(n-1)}{2}p^{n-2}q^2 + \dots + \frac{n!}{k!(n-k)!}p^{n-k}q^k + \dots + \frac{n(n-1)}{2}p^2q^{n-2} + npq^{n-1} + q^n = 1.0$$

The coefficient in front of each case → the number of different ways that the case may be observed.

Risk Analysis

$$(p+q)^n = p^n + np^{n-1}q + \frac{n(n-1)}{2}p^{n-2}q^2 + \dots + \frac{n!}{k!(n-k)!}p^{n-k}q^k + \dots + \frac{n(n-1)}{2}p^2q^{n-2} + npq^{n-1} + q^n = 1.0$$

1 2 3 4 5 6 7

- The risk of the encountering a wave with T_r year return period for a structure with n years of economic life is:

$$\text{Risk} = 1 - q^n = 1 - (1-p)^n = 1 - \left(1 - \frac{1}{T_r}\right)^n$$

- The probability that the design wave occurs exactly once in n years is:

$$\text{Prob.} = npq^{n-1} = n\left(\frac{1}{T_r}\right)\left(1 - \frac{1}{T_r}\right)^{n-1}$$

- The probability that a T_r year wave is equalled or exceeded at least twice in n years is:

$$\text{Prob.} = 1 - q^n - npq^{n-1} = 1 - \left(1 - \frac{1}{T_r}\right)^n - n\left(\frac{1}{T_r}\right)\left(1 - \frac{1}{T_r}\right)^{n-1}$$

- Exactly, only
- At least, (___ or more) More than ___
- At most, (___ or less) Less than ___

21

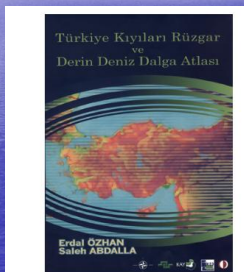
Encounter probability

- The probability that the structure will experience storm waves greater than the design condition during L years

$$\text{Risk} = 1 - q^n = 1 - (1-p)^n = 1 - \left(1 - \frac{1}{T_r}\right)^n \quad P = 1/T_r \quad P_L = 1 - \left(1 - \frac{1}{R}\right)^L$$

- $L=50, R=50, P=0.635$
- $L=50, R=100, P=0.395$

WIND AND DEEP WATER WAVE ATLAS FOR TURKISH COASTS



WIND AND DEEP WATER WAVE ATLAS FOR TURKISH COASTS

Data source

1. Wind parameters by ECMWF (European Centre for Medium range Weather Forecast). Used for 8 years (1991-1999)
2. Wind measurements (hourly averaged wind speed and direction, daily maximum) by Turkish State Meteorological Office
3. Synoptic maps (pressure) were digitized wind field at the high level of atmosphere wind field at 10m height by gradient wind model (Almost 20 years)

WIND AND DEEP WATER WAVE ATLAS FOR TURKISH COASTS

Wave modelling (hindcasting by a numerical model)

- WAM (a detailed description of physics of the wave development)
- Change in wave spectrum, H , θ , f_p

Verification

- Wave measurement in 10 stations
- ECMWF data OK
- Synoptic maps, errors were corrected

WIND AND DEEP WATER WAVE ATLAS FOR TURKISH COASTS

- Wind and wave statistics

	Long term	Extreme (Gumbel)
Black sea	8 years (91-99)	20 years (76-95)
Mediterranean sea	8 years (91-99)	17 years (79-95)
Marmara sea	4 years (91-94)	Big storms in 4 years

WIND AND DEEP WATER WAVE ATLAS FOR TURKISH COASTS

